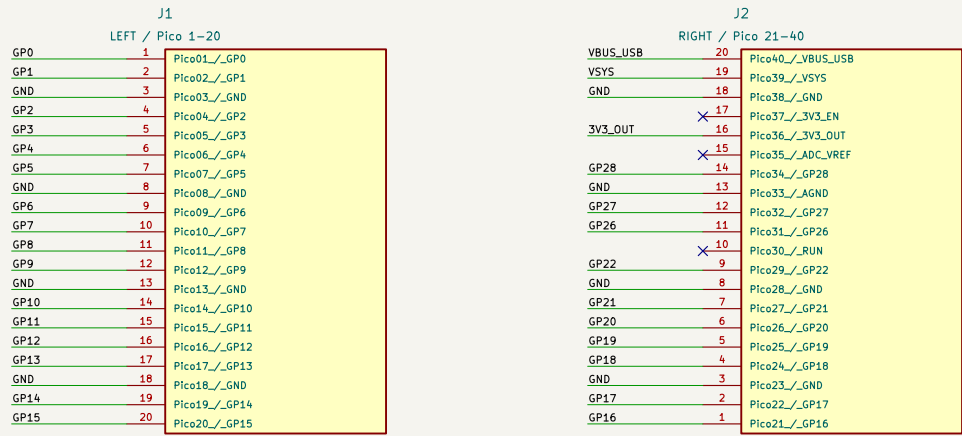


MAKE MUSIC – PROVISIONAL PICO CARRIER

Editable KiCad 10 schematic | Module order, parts, power and mechanical fit require primary integration

REMOVABLE PICO: TOP VIEW, USB AT TOP

Two 1x20 female sockets: 2.54 mm pitch, 17.78 mm row spacing. Carrier contacts are passive.



J1 pin 1 = Pico 1 (USB end). J2 pin 1 = Pico 21 (opposite USB).

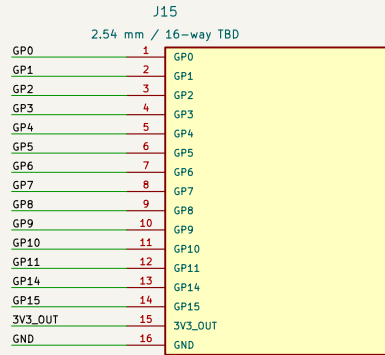
J2 pad 20 = Pico 40 at USB end: rotate J2 footprint 180 degrees relative to J1 in placement.

NC: Pico 30 RUN, 35 ADC_VREF, 37 3V3_EN. On-module defaults retained.

Pico 33 AGND tied to GND for digital touch use. GP23–25/29 are internal, not expansion.

SSQ candidate: 1.02 mm drills / 1.52 mm pads. Verify finished holes and male–header fit.

FREE GPIO EXPANSION – ORDER PROVISIONAL



GP0/GP1: optional I2C SDA/SCL.

No I2C pull–ups fitted until bus/device choice.

REVIEW ONLY – NOT FOR FABRICATION

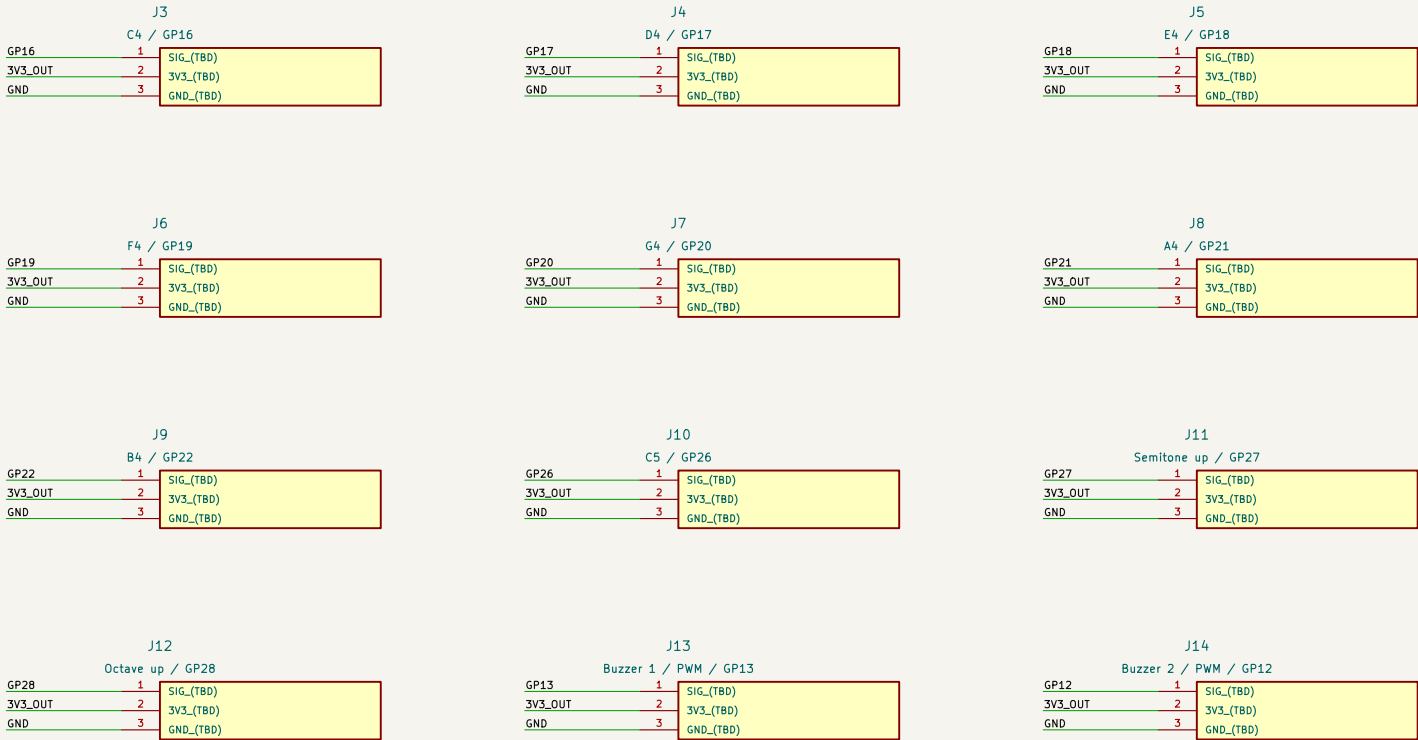
ERC validates carrier connectivity only; off–board electronics and unresolved power design are not modeled.

Sources: Raspberry Pi Pico/Pico 2 datasheets, Figure 2 pin allocation; see ../reviews/schematic.md.

TEN TOUCH / TWO BUZZER MODULE INTERFACES

ALL J3–J14: assumed 1=signal, 2=3V3, 3=GND. Verify actual module pin order before assembly.

Footprints unassigned. Only 3.3 V–compatible modules with suitable output/driver circuits may be used.



SIGNAL / POWER LIMITS TO RESOLVE

Touch: TTP223 at 3.3 V, direct mode (TOG=0). Firmware may adapt to confirmed active–high/low.

Buzzers: signal is a PWM logic input to a compatible driver module. Do not drive bare coils directly.

GP12 + GP13 share a PWM slice/frequency. Baseline firmware plays both at the same pitch.

Budget Pico 3V3_OUT current including 10 touch modules, 2 buzzer drivers and expansion loads.

J1/J2: SSQ–120–01–G–5 candidate assigned: mating fit, USB/BOOTSEL and battery clearances TBD.

Reference may be active–low (AHLB=1). Confirm actual polarity/pin order before firmware changes.

Review only – parts and power unresolved

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File: make_music.kicad_sch

Title: Make Music – provisional Pico carrier

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